

Article

Biodiversity and Resilience to Tsunamis in Chilean Urban Areas: The Role of Ecoinformatics

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Abstract: By definition, a smart city must improve its readiness for extreme events in order to confront the growing unpredictability of natural disasters. Doing this implies planning for resilience. That is, to enhance our capacity to cope, mitigate, adapt, and rebuild human settlements after a catastrophic event. Although scholars have argued that biodiversity can enhance resilience, there is a dearth of empirical research that specifically addresses this crucial issue. This research analyzes Nature's Contributions to People related to tsunami resilience. Then, the relationship between biodiversity and community resilience indexes is examined for 50 coastal Chilean cities that are prone to tsunamis, using biodiversity data from an open access database. The resilience index "population living in the first kilometer from the shoreline" was found to be correlated with species richness ($p = 0.48$) and the evenness biodiversity index, Pielou ($p = -0.47$). These results suggest that biodiversity data availability is crucial for understanding nature's contribution to human settlement resilience. Although this study was hindered by limited data availability, the potential use in other contexts remains valuable for the development of smart cities. The study highlights the need for increased biodiversity data collection on a national scale and emphasizes the use of ecoinformatics to create smart cities that can effectively respond to climate uncertainty in coastal urban areas.

Keywords: biodiversity; community resilience; tsunami; nature's contributions to people; smart city; ecoinformatics



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1. Introduction

Biodiversity encompasses every scale within the complex hierarchy of life and is the leading outcome of ecologic and evolutionary processes. As such, biodiversity may be summarized as the genetic, morphologic, and demographic expressions of life that includes the large variety of interactions among organisms and ecosystems across the landscape [1–3].

Given that biodiversity provides renewable services to humanity, now there is broad consensus on the importance of understanding the mechanisms underlying its ecological processes [4–6]. For example, biodiversity has shaped human resilience to extreme events at both the local (e.g., land use and land cover changes [7]) and regional levels (e.g., climate change [8]). This indicates that a more comprehensive and multi-level paradigm is required. New approaches to resilience must include the information required to accurately describe the relationship between biodiversity and the resilience of human societies.

We also know that the services provided by biodiversity have proven to be extremely relevant for the mitigation and adaptation of extreme events currently impacting our changing world (e.g., regulating services such as ecosystems buffers against natural disasters,

carbon sequestration, regulation of water flow [9,10]). However, listing these services has proven to be difficult since a simple quantification of species richness is not sufficient to describe biodiversity [5,11,12].

While the relationship between human and natural systems has usually been addressed through the concept of ecosystem services [13], the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) has recently proposed the Nature's Contributions to People (NCP) as a complementary approach to understand the relationship between humans and the non-human world [14]. The premise is that this framework can help deepen our understanding of how nature effectively contributes to human well-being [15]. This concept challenges the idea of an intensified human use of natural elements and acknowledges the importance of natural systems' ability to adapt to changes. NCPs have emerged as a crucial framework to foster the sustainable use of ecosystem services in which biodiversity plays a major role. For example, it is well established that livelihood declines—in terms of food and cultural supply—is closely related to biodiversity loss [13,16].

Particularly, community resilience in social systems—as used in this study—reflects the capacity of human community members and their resources to prepare, mitigate, cope, and adapt using their nearby resources to overcome uncertain and unpredictable environments [17,18]. Accordingly, the environmental dimension of community resilience plays a critical role. This dimension has been addressed regarding the ecosystem services provided by natural resources after a disaster [19,20]. For example, wetlands and grasslands can provide elements that meet basic needs—including water and food—after a disaster, while dune systems and coastal forests can act as barriers in the face of tsunamis [21,22]. On the other hand, certain authors suggest that environmental resilience is related to hazard frequency and intensity, and that this resilience is impacted by factors such as biodiversity [23,24]. Indeed, biologically diverse systems should offer a larger range of natural resources that may, in turn, contribute to resilience by means of the multiple functions provided across the landscape [25].

Despite recent efforts to describe specific biodiversity roles in the community resilience of social systems in the context of disaster prevention [19], further research is still needed. One of such efforts includes delving deeper into the characterization of the relationship between diversity and NCPs in the context of data acquisition, storage, and analysis [26]. For instance, given the increased interactions among different fields of ecological research, new disciplines (i.e., ecoinformatics and bioinformatics) require data collection, storage, and analysis [27]. Although there have been major technological advances and data repository capacity has increased during past years, the collection of biodiversity data is still a challenge [28]. This may be related to the fact that biodiversity monitoring is mainly restricted by (i) incomplete taxonomic and spatial coverage; (ii) datasets often collected under different methodologies making them inconsistent; and (iii) insufficient integration at different scales [29,30]. These drawbacks are relevant for the future of disaster resilient communities, which use technology to collect and analyze large amounts of diverse data (e.g., [31]). Consequently, questions have emerged on whether cities can be turned into "smart cities" to better respond to disasters [32,33]. Evolving smart cities concept may assist through the provision of data on biodiversity through the use of technology and digital tools such as Artificial Intelligence, cloud computing, Big Data, and Internet of Things to improve quality of life and achieve sustainable development [34,35]. To make effective decisions, the mere access to biodiversity data is not enough; the complexity of this information must also be considered. A city with extensive data collection capabilities is not necessarily resilient [36]. Hence, collecting and producing biodiversity data in a smart city should not threaten the supply of nature-based services.

This case study will focus on Chile's extremely lengthy coast (4200 km), which hosts a variety of ecosystems and different urban amenities exposed to tsunamis. High endemism and pristine ecosystems set Chile as a biogeographic island and one of the world's 36 biodiversity hotspots [37–39]. However, planning policies have so far facilitated the urban

development of coastal areas promoting activities that disregard the resilience of social communities [40].

So far, these policies have fostered urban expansion on a coastline with high ecological [41] promoting dense urban developments as by-products of private real estate agents' actions [42]. The process has increased the vulnerability of coastal areas, thwarting the population's capacity to mitigate and adapt to tsunami threats [43]; all of this is compounded by the known fact that tsunamis have increased their frequency in recent decades [44]. For example, the 2010 tsunami wreaked havoc on 600 km of the Chilean coast; destroyed 18 cities; impacted six regions; and left a toll of 181 human lives (a third of which were caused by both by the earthquake and the tsunami). Sixteen points in coastal roads recorded scour damage, approximately 17,000 homes were damaged, and 41 major coastal infrastructures were destroyed.

The main purpose of this study is to provide new insights into the relationship between biodiversity and the resilience of communities prone to tsunamis. We particularly focus on the role of ecoinformatics in informing and classifying the requirements for community resilience. We evaluate the relationship between resilience indicators and biodiversity indexes for 53 coastal Chilean cities, and characterize NCPs related to tsunami resilience. Additionally, we suggest an analysis framework for information stored in the Global Biodiversity Information Facility (GBIF) [45] aimed at measuring the biodiversity of coastal urban areas prone to tsunamis in Chile.

2. Materials and Methods

2.1. Nature's Contributions to People

A qualitative analysis of Nature's Contributions to People (NCP) was performed using the IPBES Global Assessment Report on Biodiversity and Ecosystem Services [16]. We determined whether NCPs are related to (i) tsunami resilience and (ii) the biodiversity of coastal areas. The latter was done considering each definition of the declared indicators, and proxies, described in the report. We then categorized the relationship between each NCP to both tsunami resilience and coastal biodiversity as following:

- **Related:** The NCP has a relationship with tsunami resilience (or biodiversity in coastal area) due to an explicit or implicit mention in the NCP definition, indicators or proxies to measure the NCP. For example, as tsunamis are natural hazards, NCP 9 "Regulation of Hazards and Extreme Events" is clearly related to tsunami resilience.
- **Fairly related:** We could not state or deny with any level of certainty the relationship between the NCP and tsunami resilience (or biodiversity in coastal area). This presumably happens because the structure or the scale of causality is too complex to validate the relationship. For example, it is established, but not completely, that vector-borne diseases have increased in the last 30 years and are regionally variable, i.e., NCP 10 "Regulation of Hazards and Extreme Events"; Biodiversity loss could decrease or increase disease transmissions, however, preserving endemic biodiversity should generally reduce the prevalence of diseases, which affect human health and economic security. Hence, despite the apparent biodiversity contribution to the regulation of pests and diseases, there is still not enough evidence as to how they influence the transmission of harmful organisms, particularly within coastal ecosystems [16,46,47].
- **Not related:** The NCP does not show a relationship with tsunami resilience (or biodiversity in coastal area) because there is no explicit or implicit mention have been made in the NCP definition, indicators or proxies to measure the NCP. For example, NCP 11 "Energy" was not categorized as related to tsunami resilience. Even though energy provision is essential to human wellbeing and, for instance, human resilience, no indicator, proxy, or reference to coastal hazards was found.

Finally, we validate our characterization with experts in geography, architecture, design, and disaster resilience through surveys.

2.2. Biodiversity and Resilience Relationship

2.2.1. Urban Areas

We selected 53 coastal cities known to be under tsunami threat according to the Tsunami Flood Charts (CITSU), a map that shows flooded areas and topographical surface of Chilean cities developed by the Hydrographic and Oceanographic Service of the Chilean Navy (SHOA) [48].

We used the official city definition provided by the National Institute of Statistics of Chile (INE). This definition classifies urban entities as human settlements with infrastructural continuity having a population larger than 2000 inhabitants, or having a population larger than 1001 where less than 50% of the population declares to work in a primary economic activity. Population size and economic data were collected at the census block level as provided by the latest INE Census of 2017 [49].

2.2.2. Biodiversity Data

To ascribe biodiversity information to urban areas, a convex hull polygon served as a base to build a 1 km buffer in which species occurrences were downloaded. Such information served as the database of urban biodiversity and resilience attributes (see Appendix A).

Species occurrence records were obtained from the Global Biodiversity Information Facility (GBIF) repository using the official API [50] and the pygbif library for Python [51]. GBIF is currently the leading world depository of biodiversity information with contributions from the scientific community, governmental agencies, non governmental organizations, and citizen science projects. It was conceived as a joint effort to catalog, store, and distribute all data related to species occurrences globally and currently offers more than 1.62 billion occurrence records [52]. Chile has recently subscribed as a node in GBIF [53]. As data in GBIF have increased substantially in recent years, we evaluated all occurrences in our area of interest for the last 6 years noting a peak of occurrences in 2021 (Figure A1 in Appendix B). Consequently, we restricted this analysis to the year 2021 in order to maximize information and remove potential biases associated with sample size.

For each urban area, we extracted all occurrences using the occurrence.download module of the GBIF API and removed urban areas having less than 20 occurrences.

Downloaded data was labeled according to the sources and institutions providing this information, i.e., iNaturalist, eBird from The Cornell Lab of Ornithology (CLO), World Register of Marine Species (WoRMS), and Happywhale. GBIF classifies data sources as: (i) human observation, (ii) material sample (i.e., specimen or samples taken from the field observation), (iii) material citation (i.e., reference to specimens contained in scholarly publications), (iv) machine observation (i.e., records made by a machine or sensor), (v) living specimen (i.e., observations of live specimens, as plants in gardens or animals in zoo), (vi) preserved specimen (i.e., records of specimens preserved in a collection, usually in museums), (vii) fossil specimen (i.e., records of preserved fossil specimens). To later assess any potential biases among available data, biodiversity data was classified by taxonomic groups and cities by the number of occurrences.

Three indices were used to quantify biodiversity in each area: Species richness, Gini-Simpson, and Pielou diversity [54]. Species richness (S) is not usually considered a diversity index as it simply accounts for the number of species in the location of interest [55]. We however consider it here as it provides the most simple estimation of species diversity. This was calculated by summing the number of species per city. The Gini-Simpson diversity index (D) is usually employed to account for species dominance (Equation (1)). Because every species in a location of interest might occur with a different frequency, this index weights in such asymmetry favoring whether a small number of species dominate the frequencies of occurrences [56–58].

$$D = 1 - \sum_{i=1}^S p_i^2 \quad (1)$$

where p_i is the proportional abundance of the i th species with respect to the total abundance, and S species richness.

Note that D tends to 1 when species richness is high and the number of species is equally distributed (e.g., one individual per species). Conversely, D will be 0 when most records belong to the same specie. Pielou index (J) (Equation (3)) evaluates equitability, i.e., evenness (Equation (3)) [59].

$$H' = - \sum_{i=1}^S p_i \cdot \ln(p_i) \quad (2)$$

$$J = \frac{H'}{H'_{max}} \quad (3)$$

where H' is Shannon-Wiener index, and H'_{max} the maximum H' value known for the site [54].

We studied differences among the data distribution of biodiversity indices graphically using violin plots and compared cities with more than 20 and 100 occurrences.

2.2.3. Resilience of Social Communities Data

A resilience index was built considering variables from the economic, physical, and environmental resilience dimensions. The economic dimension includes the monthly incomes related to fishery and tourism. Income is commonly used to measure communities' coping capacity after a disaster [60,61]. More affluent individuals, i.e., with larger incomes, usually have better chances to satisfy their basic needs after a disaster such as accessing temporary shelter, for instance. To construct this variable we took the monthly income from the total number of people with incomes from the fishing and tourism industries. Economic data was obtained from the National Socioeconomic Characterization survey [62].

We addressed the physical dimension of community resilience evaluating aspects of urban morphology. The way in which urbanization has been developed on the Chilean coast responds to economic demands of the private, real estate, and tourism industries. Accordingly, planning instruments encourage urban expansion on the 1st km of the coastline, which usually is the area of the largest ecological value [41]. Consequently, such urban expansion fosters dense urbanization's that may thwart adaptations of social communities against tsunami hazards [43]. We used population density in the first coastal kilometer as a measure of urban morphology assuming that higher population density will be associated to less community resilience. We calculated this variable using the 2017 Census [49].

Finally, the environmental dimension was measured using the natural tsunami mitigation area for each urban area. Dune systems with vegetation and coastal forest can reduce the destructive force of tsunamis, while wetlands with flood zones, or the percentage wetland area, can act as a buffer against bigger floods caused by tsunamis [19,63]. Hence, we computed the total amount of tsunami flooded area covered by these natural systems for each city.

2.2.4. Biodiversity and Resilience Relation Analysis

A ranked pairwise correlation between each biodiversity index (S , D , and J) and resilience indices (i.e., incomes, population, and mitigation areas) was used to describe their relationship.

3. Results

3.1. Nature's Contributions to People

Results show that 11 out of 18 NCPs (61%) are related to resilience to tsunamis. Besides, 13 out of 18 NCPs (72%) are related to biodiversity in coastal areas, and in contrast to the relation with tsunami resilience, we could not find NCPs purely not related to coastal biodiversity (Table 1). See Appendix C for a detailed account of the data and survey results.

Table 1. Nature’s Contribution to People and revision check according to their relation to tsunami resilience and biodiversity in coastal areas.

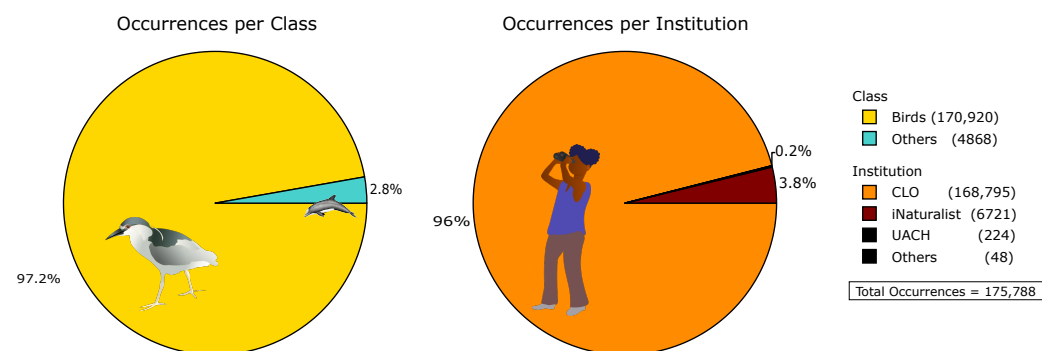
NCP	Is It Related to Tsunami Resilience?	Is It Related to Biodiversity (In Coastal Areas)?
(1) Habitat Creation and Maintenance	✓	✓
(2) Pollination and Dispersal of Seeds	×	✓
(3) Regulation of Air Quality	×	0
(4) Regulation of Climates	0	✓
(5) Regulation of Ocean Acidification	0	✓
(6) Regulation of Freshwater Quantity, Location, and Timing	✓	✓
(7) Regulation of Freshwater Quality	✓	✓
(8) Formation, Protection, and Decontamination of Soils	✓	✓
(9) Regulation of Hazards and Extreme Events	✓	✓
(10) Regulation of Organisms Detrimental to Humans	0	0
(11) Energy	×	0
(12) Food and Feed	✓	✓
(13) Materials and Assistance	✓	0
(14) Medicinal, Biochemical, and Genetic Resources	0	✓
(15) Learning and Inspiration	✓	✓
(16) Physical and Psychological Experiences	✓	✓
(17) Supporting Identities	✓	0
(18) Maintenance of Options	✓	✓

✓ = Related, 0 = Fairly related, × = Not related.

3.2. Biodiversity

We removed three cities having less than 20 occurrences: Tocopilla (7), Hanga Roa (8), and Iloca (16). These outliers show that the number of occurrences seems to strongly influence the results.

The total number of occurrences found across the selected cities is 175,788 (Figure 1). Arica, the northernmost city, had the highest amount of data (15,091 occurrences) and Lebu had the lowest (21). Most records (99.9%) come from human observations of publically available platforms such as eBird (CLO) and iNaturalist. Of these, 97.2% of occurrences belong to the class *Aves*, followed by five classes of vascular plants in the phylum *Tracheophyta*. These are mainly represented, given their number of occurrences, by the classes *Magnoliopsida* and *Liliopsida*, as detailed in Appendix D.

**Figure 1.** Biodiversity occurrences for 50 cities in 2021. The left panel shows the data proportion by taxonomic class rank. The right panel is the data proportion contributed by institutions.

Biodiversity Indices

The average species richness (S) for the 50 cities is 152. The mean value of the Gini-Simpson index (D) is 0.97 and the mean value of the Pielou index (J) is 0.86. Values per city are shown in Figure 2A and occurrence distribution is presented in Figure 2B. Detailed results per index and their distribution are presented below.

- **Species Richness:** As shown in (Figure 2C), Lebu and Coquimbo show the lowest and largest species richness with 16 and 364 respectively. A consistent trend of increasing species richness at mid-latitudes echoes the known biogeographic gradient observed throughout Chile with a peak in the central region of Chile [64]. Urban areas between the Coquimbo region (La Serena and Coquimbo) and the Biobío region (Talcahuano, Penco, and San Pedro de la Paz) host the largest values for species richness.
- **Gini-Simpson:** The lowest D value (0.93) belongs to Lebu, while the cities having the largest D values (0.98) are Penco, Pichilemu, and Talcahuano. Most cities show large values for D and, thus, exhibit a high diversity and low dominance. Three cities have a D index below 0.95 and also share a low number of occurrences. These are Carelmapu, Quidico, and Lebu with 40, 25, and 21 occurrences each (Figure 2D).
- **Pielou:** The lowest J value (0.679) occurs in Viña del Mar and the maximum J value (0.984) is seen in Carelmapu. Six cities have the largest values for J (>0.95). These are Chañaral, Caleta Tumbes, Lebu, Quidico, Tirúa, and Carelmapu; which coincide with the six cities with the lowest occurrences of 46, 98, 21, 25, 88, and 40 (Figure 2E). This led us to analyze the evolution of the index while excluding cities with the lowest value for J.
- **Biodiversity Indices Distribution:** To evaluate whether occurrences per city affect the statistical distribution of biodiversity indices, we produced violin plots for two groups of cities based on the number of occurrences. About 50 cities had less than 100 occurrences, while 44 cities showed a larger number of occurrences. The distribution of S approximated a slightly skewed distribution for both groups of cities; most cities have D close to 1; J showed a unimodal distribution resembling a normal distribution. After excluding cities with less than 100 occurrences, the mean J value decreased from 0.86 to 0.84. However, the frequency distribution preserved its shape with most cities showing J values in the [0.8, 0.9] range (Figure 3).

3.3. Community Resilience

Social community resilience was evaluated using the economic (Figure 4), physical, and environment (Figure 5) dimensions proposed in the CORE community resilience model proposed by [19]. (see further details in Appendix D).

Economic Dimension

The Chilean national survey of socioeconomic conditions of 2020 [62] allowed us to obtain the average value of monthly per capita income for tourism and fishery related items (i.e., 305,000 CLP). The cities with high incomes are Mejillones (>1,500,000 CLP/month), San Pedro de la Paz (>1,200,000 CLP/month) and Viña del Mar (>980,000 CLP/month). At the same time, 8 out of 50 cities do not report fishery and tourism incomes: Papudo, Quintero, Santo Domingo, Constitución, Cobquecura, Puerto Saavedra, Queule, and Bahía Mansa.

Physical Dimension

The mean population size in the first kilometer from the coastline for the 50 studied cities is approximately 22,000. The cities that present the most population in this coastal zone are Antofagasta (1.6×10^5), Iquique (1.1×10^5), La Serena (10^5), and Valparaíso (9×10^4). On the other hand, Hualpén and Maullín show very low population sizes. Also, it is not possible to see any trend through latitude, however, three of the most populated cities are at lower latitudes.

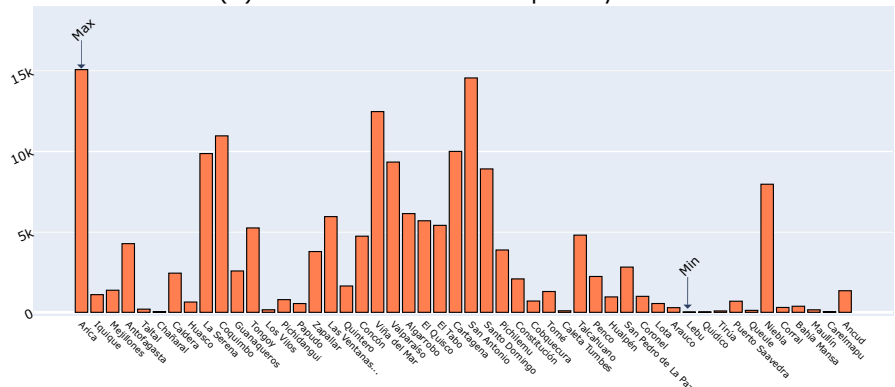
Environmental Dimension

The mitigation area mean value for the 50 cities is approximately 0.8 km². Talcahuano is the city with the largest mitigation area (4.8 km²) followed by La Serena (4.1 km²). Fourteen cities do not present mitigation areas.

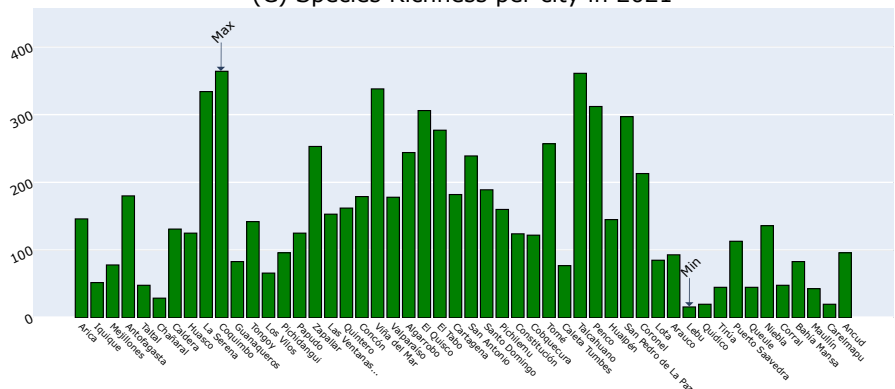
(A) Table of BD indexes

City	Occurrences	Species-Richness	Gini-Simpson	Pielou
Arica	15091	146	0.98	0.81
Iquique	1100	52	0.95	0.84
Mejillones	1374	78	0.97	0.85
Antofagasta	4270	180	0.97	0.76
Taltal	197	48	0.96	0.92
Chañaral	46	29	0.96	0.98
Caldera	2436	131	0.97	0.82
Huasco	639	125	0.98	0.89
La Serena	9871	334	0.98	0.76
Coquimbo	10,973	364	0.98	0.76
Guanqueros	2572	83	0.97	0.81
Tongoy	5247	142	0.97	0.82
Los Vilos	159	66	0.98	0.95
Pichidanguí	790	96	0.98	0.89
Papudo	546	125	0.98	0.92
Zapallar	3779	253	0.98	0.81
Las Ventanas Horcón Mañencillo	5951	153	0.98	0.85
Quintero	1637	162	0.98	0.88
Concón	4734	179	0.98	0.83
Viña del Mar	12,481	338	0.97	0.68
Valparaíso	9342	178	0.96	0.72
Algarrobo	6139	244	0.98	0.79
El Quisco	5693	306	0.98	0.75
El Tabo	5408	277	0.98	0.79
Cartagena	10,006	182	0.98	0.81
San Antonio	14,573	239	0.98	0.82
Santo Domingo	8917	189	0.98	0.85
Pichilemu	3876	160	0.99	0.89
Constitución	2079	124	0.98	0.86
Cobquecura	703	122	0.98	0.88
Tomé	1295	257	0.98	0.85
Caleta Tumbes	98	77	0.98	0.98
Talcahuano	4800	361	0.99	0.80
Penco	2229	312	0.99	0.84
Hualpén	959	145	0.98	0.85
San Pedro de La Paz	2813	297	0.98	0.80
Coronel	994	213	0.98	0.87
Lota	548	85	0.97	0.86
Arauco	291	93	0.98	0.91
Lebu	21	16	0.93	0.98
Quidico	25	20	0.94	0.98
Tirúa	88	45	0.97	0.97
Puerto Saavedra	691	113	0.98	0.89
Queule	122	45	0.97	0.95
Niebla	7965	136	0.97	0.80
Corral	303	48	0.95	0.85
Bahía Mansa	378	83	0.97	0.90
Mauñín	156	43	0.96	0.91
Caremapu	40	20	0.94	0.98
Ancud	1343	96	0.98	0.90

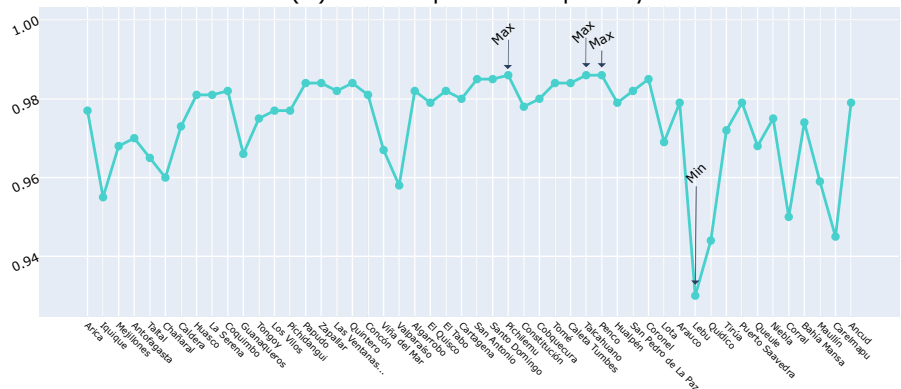
(B) Number of occurrences per city in 2021



(C) Species Richness per city in 2021



(D) Gini-Simpson index per city



(E) Pielou index per city

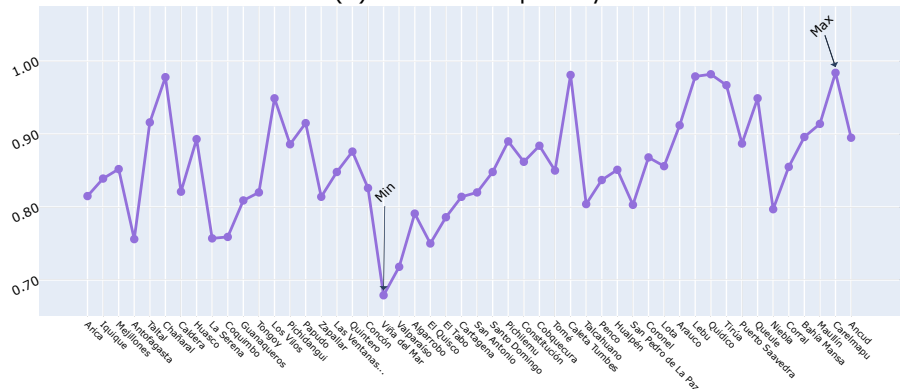


Figure 2. Biodiversity indices for 50 cities in 2021. Cities are sorted by latitude from north to south.

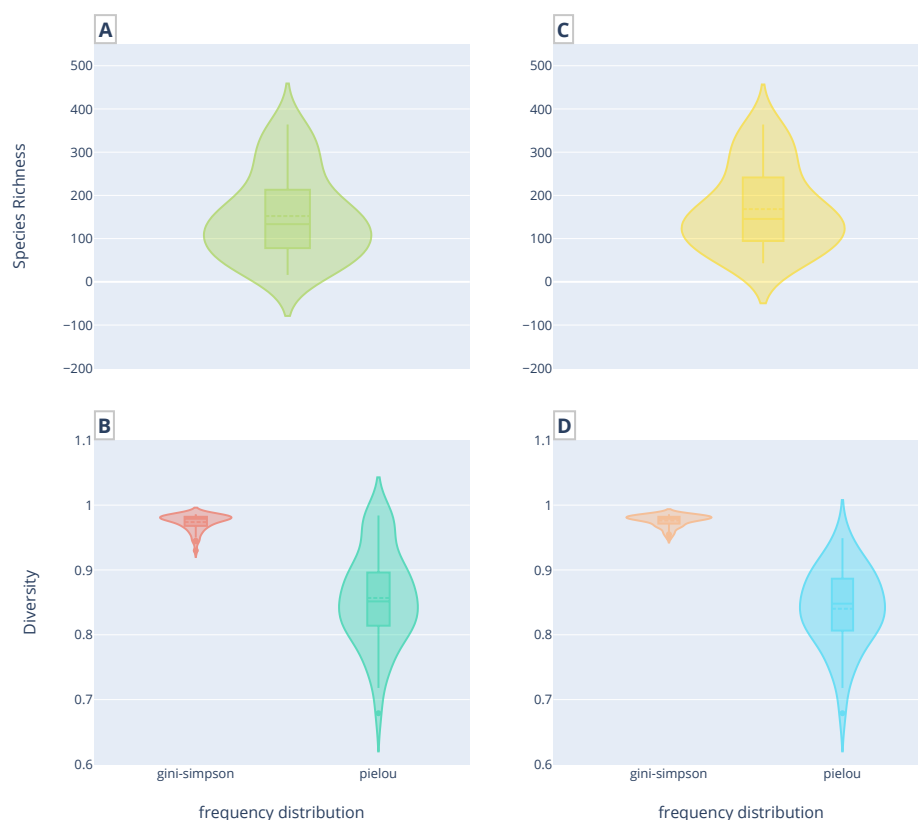


Figure 3. Biodiversity indices distribution comparison between the numbers of cities used. Plots (A,C) correspond to the distributions of species richness (S), while plots (B,D) show the distributions of Gini-Simpson (D) and Pielou (J) indices. Note that plots (A,B) use data from the 50 cities with a relatively small number of occurrences (i.e., <100), while plots (C,D) use data from the 44 cities with a larger number of occurrences (i.e., >100).

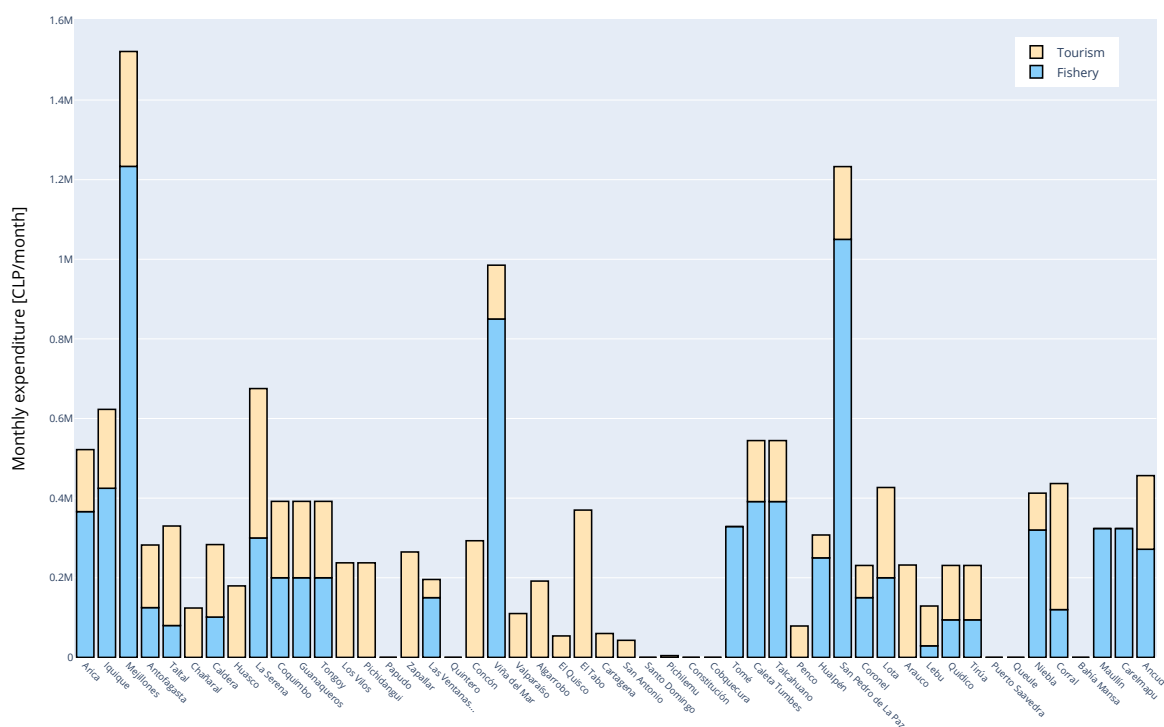


Figure 4. Monthly expenditures of economic dimensions related to the social community resilience for 50 coastal cities in Chile. Note that cities are sorted in a latitudinal gradient from north to south.

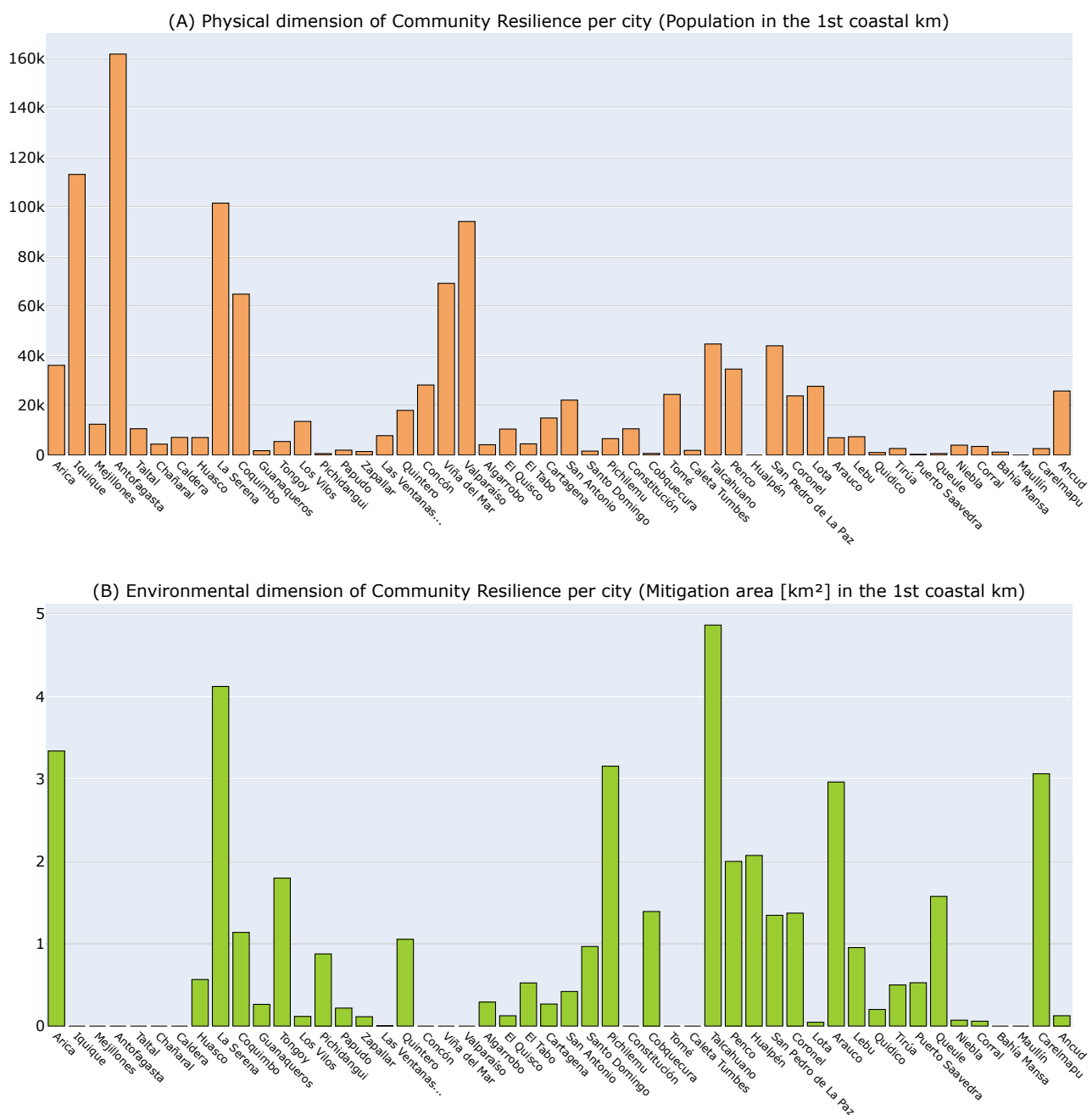


Figure 5. Physical and environmental dimension of social community resilience for the 50 cities studied in this research sorted in a north to south orientation.

3.4. Biodiversity and Resilience Relation Analysis

We studied the ranked Spearman correlation between biodiversity indices and indices of social community resilience (Figure 6). While no strong correlations are observed, species richness seems to be mostly associated with the number of people in the first kilometer of the coast and the designated mitigation areas. Income by fishery does not seem to be associated to any biodiversity index, nor S. Interestingly, Tourism, while showing a weak correlation is still significant considering $p < 0.1$. The coefficient correlation between Gini-Simpson index and the indices describing the economic dimension of resilience (i.e., both tourism and fishery incomes) are negative. Finally, there is no correlation between the Pielou index and the environmental dimension of social community resilience such as the extent of mitigation areas, while a negative relationship exists with all other three social resilience indices.

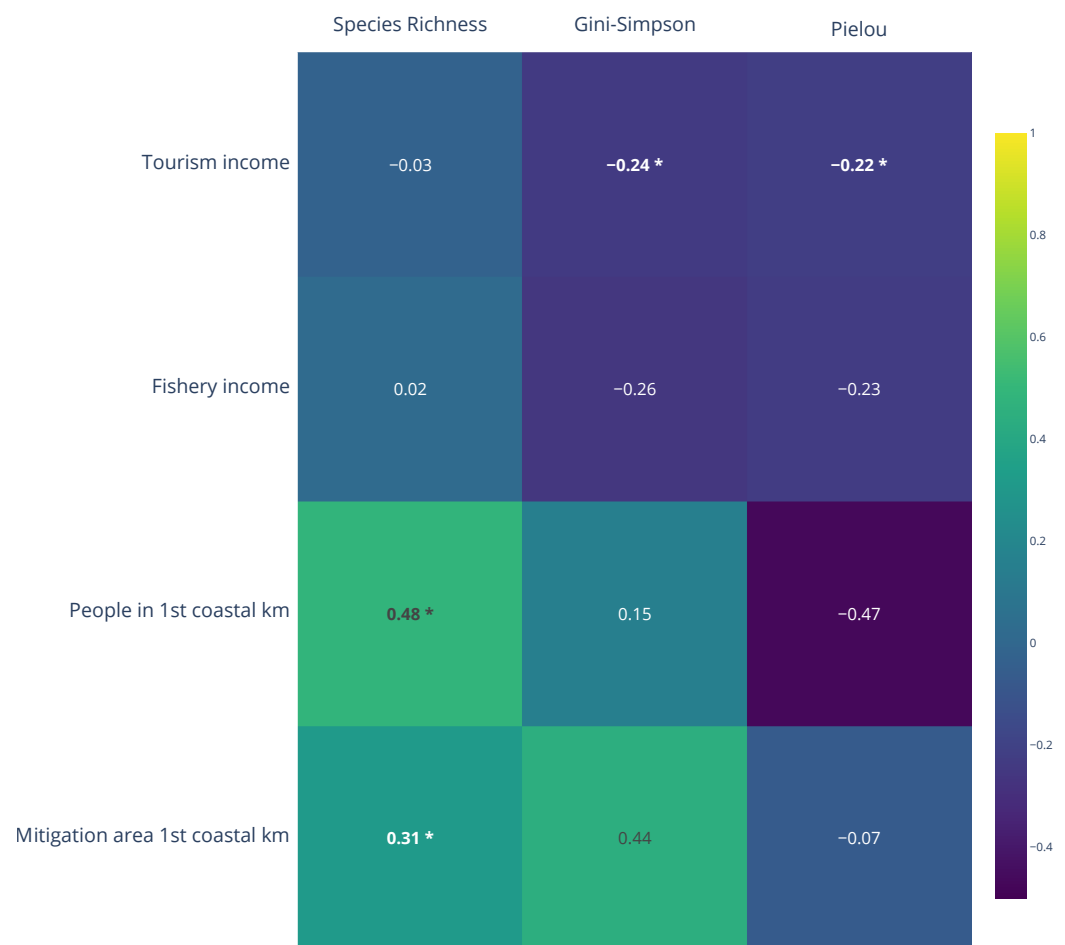


Figure 6. Spearman ranked correlations between biodiversity indices and resilience indices. Note that incomes are reported per capita. (* indicates $p < 0.1$, all other relationships are not significant).

4. Discussion

NCP approaches are useful to understand the relationship between humans and nature under the premise of preserving a biodiversity that may foster resilience among social and natural systems. From the 18 NCPs defined by IPBES, 11 of them were found to contribute to the analysis on tsunami catastrophe resilience in Chile. Despite the growing interest in nature-based solutions to mitigate hazards, the impact of anthropogenic changes on shoreline landscapes and their biodiversity is still not fully understood [16].

Our analysis of biodiversity data highlights important biases regarding available information on species in coastal cities prone to tsunami disturbances. While good descriptions on Chilean biota exist [65–68], such bias could be partly explained by the limited availability of georeferenced biodiversity information in ecoinformatics databases along the Chilean coast. Here, we document such information sources and show that most data comes from research institutions and science-related community initiatives (Figure 1), revealing how sampling schemes might be largely incidental and could result in taxonomic, geographic, and temporal biases [28,69]. For instance, we show that most observations are related to birds (i.e., 97.2% of the occurrences reported belong to class *Aves*); this clearly misses how biodiversity contributes to tsunami resilience since the association between regulatory ecosystem services and bird biodiversity is not evident. These type of sources clearly bias our results. For example, the highest correlation coefficient observed is the relationship between S and the number of people living in the first coastal kilometer ($\rho = 0.48$) (Figure 6), which may be related to the type of information source, i.e., human observations. Higher human population densities are associated with increased data availability, which may contribute to greater species richness in those areas. Also, during 2021, less than

180,000 occurrences were collected in the 50 cities most exposed to tsunamis in Chile. Both results show that the lack of data and the reliance on citizen science observations can result in biases that hinder the understanding of the relationship between biodiversity and resilience to tsunamis in coastal cities. Our study seeks to contribute in exposing this data availability gap.

Additionally, an interesting association is found between the number of occurrences and species richness. For example, coastal cities with great richness may clearly be the product of larger sampling efforts spearheaded by, perhaps, important research institutions as happens in La Serena and the Biobío region. In turn, this may lead to a larger number of occurrences that translate into more species being recorded in the database. However, cities in the Biobío region do not follow to this pattern as this species-rich area is not associated with a large number of occurrences (see Figure 2). This points towards additional factors that could explain species richness in this area, including the existence of upwelling zones [70].

As shown in Figure 3, the statistical similarities of the Pielou index (J) distribution when comparing cities with small and large number of occurrences reveals how the number of occurrences impacts the value of J . None of these six cities had more than four occurrences per species (occurrences/ S), which naturally leads to a high evenness. However, despite the exclusion, biases remained: the new highest J value cities were Queule (122 occurrences and $S = 45$) and Los Vilos (159 occurrences and $S = 66$) ($J = 0.949$). Both cities have less than 10 occurrences per species, which seems unsustainable for any species population and would lead to its extinction.

Similarly, our correlation results (Figure 6) showed a low association between biodiversity and tourism and fishery-related incomes. This seems to be related to the nature of the only survey accounting for this type of activities in Chile [62]. For example, the survey reports no income from fisheries and tourism across nine of the 50 cities studied here. Similarly, 14 cities report no mitigation resources on the first kilometer inland from the coastline.

The negative relationship between the biodiversity and the resilience of urban coastal areas shown here does not support previous findings [14,71,72]. In fact, the weak, albeit significant, correlation of biodiversity with the population in the first kilometer off the shoreline suggests that we are still building on comprehensive databases that may account for the extant biodiversity in these areas. In fact, most of the occurrences in the database used (more than 99% of total records) come from citizen-science initiatives. These projects are known to be strongly biased as they are closely related to the observers' interests [28].

The availability of biological data is crucial for decision-making [68]. While this is a task part of Global Biodiversity Information Facility objectives (declared as a priority area [52]), we show that the incompleteness of the Chilean coastal dataset precludes efforts to establish the relationship between biodiversity and NCPs for community resilience.

Therefore, this gap between the need to collect biodiversity data and their actual availability could provide new opportunities for the development of smart city tools. In fact, while some counterexamples exist [73], it is widely believed that technological optimizations in the urban context should strengthen our capacity to collect biodiversity information and to understand coastal cities as social-ecological units, thereby developing 'smarter' approaches to resilience-related urban planning [35]. Recently, some efforts have been made by the scientific community in Chile to consider the role of ecoinformatics as a tool to provide updated biodiversity information and contribute to decision-making in urban planning. For example, while many databases host vegetation information describing the coastal flora in Chile (e.g.: National inventory of species of Chile [65], and the Territorial Information System [67]), and report occurrences, geographic locations or dates, additional efforts are needed, as shown in the recent report prepared by a large group of Chilean scientists [68].

The number of studies discussing the relationship between biodiversity and community resilience remains limited, and empirical evidence is scarce to date. Therefore, the value of this study lies in its methodological framework which can be applied to fu-

ture research in the area. While the approach implemented may have been hindered by scant data availability, we believe that its potential use in other contexts is relevant for the development of smart and resilient cities.

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Abbreviations

The following abbreviations are used in this manuscript:

NCP	Nature's Contribution to People
GBIF	Global Biodiversity Information Facility
S	Species richness
D	Gini-Simpson index
J	Pielou index

Appendix A. Cities' Resilience Data

The next table shows all data used for resilience index. Source data is the National Socioeconomic Characterization survey [62].

Table A1. Community resilience data for the 50 coastal urban areas studied.

City	Total Number Interviewed	Tourism Workers	Tourism Income [CLP/ Month]	Tourism Income per Capita [CLP/ Month pp]	Fishery Workers	Fishery Income [CLP/ Month]	Fishery Income per Capita month pp]	Income [CLP/ Month pp]	Per Capita Income [CLP/Month pp]	Tourism-Dependent Economy [%]	Percentage of Dependent Occupations in Sector [%]	Coastal Occupation Service	Percentage of the Community Working in the Fishery Sector [%]	People in 1 Coastal Km	Mitigation Area 1 Coastal Km [km ²]	City Longitude	City Latitude
Arica	6665	151	23,525,999	155,801	23	8,420,000	366,087		120,486	3	68		1	36,181	3	−70.31	−18.48
Iquique	5120	138	2,734,6000	198,159	25	10,620,000	424,800		184,508	3	77		1	113,178	0	−70.13	−20.22
Tocopilla	303	5	938,000	187,600	2	0	0		138,221	2	63		2	23,319	0	−70.19	−22.09
Mejillones	172	7	2,020,000	288,571	3	3,700,000	1,233,333		151,936	8	65		5	12,426	0	−70.40	−23.07
Antofagasta	4742	96	15,097,048	157,261	2	250,000	125,000		191,576	2	66		0	161,712	0	−70.48	−24.60
Taltal	150	2	500,000	250,000	4	320,000	80,000		107,367	3	59		9	10,605	0	−70.48	−25.40
Chañaral	247	5	620,000	124,000	2	0	0		113,552	2	64		2	4407	0	−70.62	−26.34
Caldera	434	10	1,820,000	182,000	15	1,520,000	101,333		116,799	4	62		10	7105	0	−70.79	−27.06
Hanga Roa	2516	54	5,946,000	110,111	7	0	0		133,727	2	70.0		1	5537	0	−109.42	−27.15
Huasco	156	12	2,155,000	179,583	0	0	0		129,487	11	69		0	7071	1	−71.22	−28.47
La Serena	2124	32	12,007,000	375,219	1	300,000	300,000		124,098	3	77		1	101,599	4	−71.24	−29.89
Coquimbo	2062	37	7,100,000	191892	6	1200,000	200,000		117,528	3	63		1	64,894	1	−71.31	−29.96
Guañaqueros	2062	37	7,100,000	191,892	6	1,200,000	200,000		117,528	3	63		1	1757	0	−71.41	−30.19
Tongoy	2062	37	7,100,000	191,892	6	1,200,000	200,000		117,528	3	63		1	5447	2	−72.50	−30.26
Los Vilos	164	4	950,000	237,500	3	0	0		119,878	5	53		5	13,567	0	−72.50	−31.91
Pichidangui	164	4	950,000	237,500	3	0	0		119,878	5	53		5	603	1	−71.52	−32.14
Papudo	117	0	0	0	0	0	0		107,634	0	62		0	1961	0	−71.44	−32.50
Zapallar	142	5	1,324,000	264,800	1	0	0		100,972	9	75		2	1414	0	−71.44	−32.58
Las Ventanas – Horcón – Maitencillo	173	7	320,000	45,714	3	450,000	150,000		75,416	2	53		5	7823	0	−71.46	−32.70
Quintero	378	0	0	0	0	0	0		76,802.0	2	64		4.0	18,005	1	−71.52	−32.79
Concón	438	10	293,0000	293,000	1	0	0		272,439	2	70		2	28,250	0	−71.49	−32.93
Viña del Mar	3790	59	7,974,000	135,153	6	5,100,000	850,000		165,553	1	70		0	69,238	0	−71.52	−33.02
Valparaíso	2516	54	5,946,000	110,111	7	0	0		133,727	2	70		1	94,191	0	−71.61	−33.04
Algarrobo	252	9	1,725,000	191,667	1	0	0		75,474	9	67		1	4,159	0	−71.65	−33.36
El Quisco	303	12	646,500	53,875	1	0	0		57,018	4	65		1	10,471	0	−71.68	−33.40
El Tabo	119	2	740,000	370,000	1	0	0		54,824	11	75		4	4510	1	−71.63	−33.45
Cartagena	285	7	420,000	60,000	0	0	0		59,388	2	67		0	14,949	0	−71.60	−33.54
San Antonio	710	7	300,000	428,57	16	0	0		88,892	0	63		6	22,163	0	−71.59	−33.59
Santo Domingo	123	2	0	0	0	0	0		102,413	0	70		0	1586	1	−72.01	−33.64
Pichilemu	269	9	40,000	4444	5	0	0		68,959	0	59		5	6605	3	−72.01	−34.39
Iloca	94	1	0	0	1	0	0		96,106	0	75		3	1655	0	−72.18	−34.94
Constitución	327	5	0	0	1	0	0		110,719	0	53		1	10,581	0	−72.41	−35.33
Cobquecura	82	1	0	0	1	0	0		70,183	0.0	56		3	664	1	−72.79	−36.13

Table A1. Cont.

City	Total Number Interviewed	Tourism Workers	Tourism Income [CLP/ Month]	Tourism Income per Capita [CLP/ Month pp]	Fishery Workers	Fishery Income [CLP/ Month]	Fishery Income per Capita month pp]	Income [CLP/ Month pp]	Per Capita Income [CLP/Month pp]	Tourism-Dependent Economy [%]	Percentage of Coastal Occupancy in the Fishery Sector [%]	Percentage of the Community Working in the Fishery Sector [%]	People in 1 Coastal Km	Mitigation Area 1 Coastal Km [km ²]	City Longitude	City Latitude
Tomé	435	5	0	0	7	2,300,000	328,571		90,267	48	67	5	24,456	0	−72.95	−36.60
Caleta Tumbes	1797	21	3,218,000	153,238	15	5,870,000	391,333		121,723	1	67	2	1880	0	−73.09	−36.93
Talcahuano	1797	21	3,218,000	153,238	15	5,870,000	391,333		121,723	1	67	2	44,793	5	−73.11	−36.74
Penco	499	13	1,025,000	78,846	3	0	0		109,109	2	73	2	34,665	2	−73.0	−36.73
Hualpén	943	14	804,000	57,429.0	2	500,000	250,000		111,036	1	69	1	0	2	−73.00	−36.79
San Pedro de La Paz	2636	35	6,397,002	182,771	6	6,300,000	1,050,000		180,898	1	78	1	44,060	1	−73.10	−36.85
Coronel	1101	12	971,000	809,17	9	1,350,000	150,000		93,459	1	61	3	23,868	1	−73.14	−36.98
Lota	464	8	1,814,000	226,750	3	600,000	200,000		79,651	5	60	2	27,737	0	−73.15	−37.09
Arauco	204	5	1,160,000	232,000	0	0	0		222,578	3	63	0	6986	3	−73.31	−37.24
Lebu	219	3	300,000	100,000	21	610,000	29,048		64,363	2	50	26	7399	1	−74.65	−37.60
Quidico	89	6	820,000	136,667	7	660,000	94,286		129,202	2	68	5	1017	0	−73.0	−38.24
Tirúa	89	6	820,000	136,667	7	660,000	94,286		129,202	2	68	5	2637	0	−73.49	−38.34
Puerto Saavedra	79	2	0	0	3	0	0		103,506	0	66	10	306	1	−73.39	−38.78
Queule	84	0	0	0	3	0	0		75,631	0	61	17	642	2	−73.20	−39.39
Niebla	3259	60	5,551,000	92,517	5	1,600,000	320,000		144,752	2	72	0	3989	0	−73.39	−39.86
Corral	100	3	950,000	316,667	5	600,000	120,000		108510	9	51	12	3469	0	−73.43	−39.88
Bahía Mansa	88	5	0	0	3	0	0		54,659	6	55	10	1170	0	−73.73	−40.58
Mauñilín	92	1	0	0	11	3,560,003	323,637		88,870	0	59	27	0	0	−73.59	−41.61
Carelmapu	92	1	0	0	11	3,560,003	323,637		88,870	0	59	27	2609	3	−73.70	−41.74
Ancud	196	3	555,000	185,000	7	1,901,000	271,571		113,878	2	67	11	25,825	0	−73.81	−41.87

Appendix B. GBIF Registers in Chile from 2017 to 2022

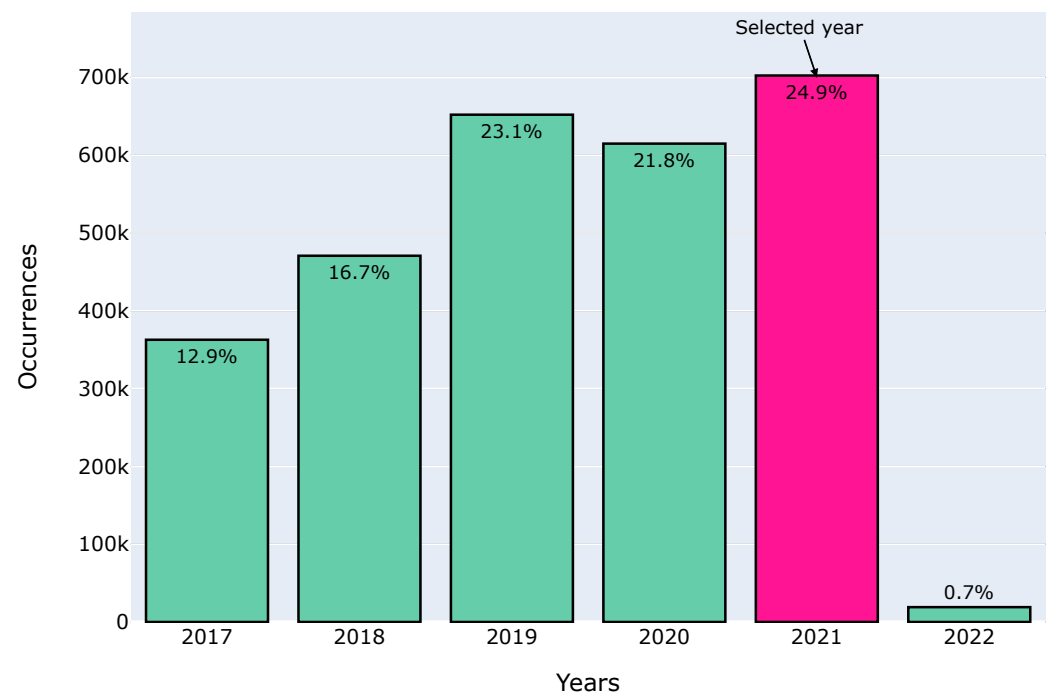


Figure A1. Biodiversity registers (occurrences) of Chile from 2017 to 2022 in the GBIF platform. The year selected to extract biodiversity data is highlighted in pink and corresponds to the year with the highest number of occurrences.

Appendix C. Nature's Contributions to People Survey

We surveyed experts in resilience, using a three parts Google Forms:

- Respondent information (not required): Name; e-mail; work area or discipline, as check list between the following areas: Anthropology, Architecture, Design, Engineering, Health (Nursing, Medicine, etc.), Law, Sciences (Biology, Chemistry, Biochemistry, Physics), Sociology or Other. The majority of those surveyed came from the area of architecture and geography.

- Definitions:

Community Resilience:

In this work, we consider resilience as community resilience That is, the adaptation capacity of cities and their inhabitants to these events, without losing their identity, structure and function.

Nature's Contributions to People:

The Contributions of Nature to People or NCP for its acronym in English (Nature's Contributions to People) are the contributions of nature to humanity. They were described by IPBES in 2017 as the way to relate non-human nature to the well-being of people, updating the notion of ecosystem services. Currently 18 NCPs are defined (described below).

NCP 1: Habitat creation and maintenance

The formation and continued production, by ecosystems, of ecological conditions necessary or favourable for living beings important to humans.

NCP 2: Pollination and dispersal of seeds

Facilitation by animals of movement of pollen among flowers, and dispersal of seeds, larvae, or spores of organisms beneficial or harmful to humans.

NCP 3: Regulation of air quality

Regulation (by impediment or facilitation) by ecosystems of atmospheric gases; filtration, fixation, degradation, or storage of pollutants.

NCP 4: Regulation of climate

Climate regulation by ecosystems (including regulation of global warming) through effects on emissions of greenhouse gases, biophysiscal feedbacks, biogenic volatile organic compounds, and aerosols.

NCP 5: Regulation of ocean acidification

Regulation, by photosynthetic organisms, of atmospheric CO₂ concentrations and so seawater pH.

NCP 6: Regulation of freshwater quantity, location and timing

Regulation, by ecosystems, of the quantity, location, and distribution of the flow of surface and groundwater used for consumption, irrigation, transportation, hydropower, and as a support for non-material contributions (NCP 15, 16, 17). Regulation of flow to water-dependent natural habitats that in turn positively or negatively affect people downstream, including through flooding (wetlands, including ponds, rivers, lakes, swamps). Modify groundwater levels, which can enhance dryland salinization in non-irrigated landscapes.

NCP 7: Regulation of freshwater and coastal water quality

Regulation, through the filtration of particles, pathogens, excess nutrients and other chemical substances, by ecosystems or particular organisms, of the quality of water used directly (for example, drinking) or indirectly (for example, aquatic foods, irrigated food and fiber crops, freshwater and coastal habitats of heritage value).

NCP 8: Formation, protection and decontamination of soils

Sediment retention and erosion control, soil formation, and maintenance of soil structure and processes (such as decomposition and nutrient cycling) that underlie the continued fertility of soils important to humans. Filtration, fixation, degradation or storage of chemical and biological contaminants (pathogens, toxics, excess nutrients) in soils and sediments that are important to humans.

NCP 9: Regulation of hazards and extreme events

Enhancement, by ecosystems, of impacts on humans or their infrastructure caused by, for example, floods, wind, storms, hurricanes, seawater intrusion, tidal waves, heat waves, tsunamis, high noise levels, fires. Reduction, by ecosystems, of hazards such as landslides, avalanches. Increase, by organism, in the probability of hazards (for example: beaver dams affecting flooding).

NCP 10: Regulation of organisms detrimental to humans

Regulation, by ecosystems or organisms, of pests, pathogens, predators, competitors, etc. affecting humans, plants and animals, including for example:

- Regulation by predators or parasites of the population size of important non-harmful animals, (e.g., large populations of herbivores by wolves or lions).
- Regulation (by impediment or facilitation) of the abundance or distribution of potentially harmful organisms (e.g., poisonous, toxic, allergenic, predators, parasites, competitors, disease vectors and reservoirs) on the land or seascape.
- Disposal of animal carcasses and human carcasses by scavengers (e.g., vultures in Zoroastrian traditions and some of Tibetan Buddhism).
- Regulation (by impediment or facilitation) of biological deterioration and infrastructure degradation (e.g., damage by pigeons, bats, termites, strangling figs in buildings).

NCP 11: Energy

Production of biomass-based fuels, such as biofuel crops, animal waste, firewood, agricultural residue.

NCP 12: Food and feed

Food production from wild, managed, or domesticated organisms, such as fish, bushmeat and edible invertebrates, beef, poultry, game, dairy products, edible crops, wild

plants, mushrooms, honey. Production of food (fodder and feed) for domestic animals (livestock, work and support animals, pets) or for aquaculture, from the same sources.

NCP 13: Materials and assistance

Production of materials derived from organisms in cultivated or wild ecosystems for construction, clothing, printing, ornamental purposes (for example: wood, fibers, waxes, paper, resins, dyes, pearls, shells, coral branches). Direct use of living organisms for decoration (ornamental plants, birds, fish in homes and public spaces), companionship (pets), transportation, and labor (grazing, searching, guiding, surveillance).

NCP 14: Medicinal, biochemical and genetic resources

Production of materials derived from organisms (plants, animals, fungi, microbes) used for medicinal and veterinary purposes. Production of genes and genetic information used for plant and animal breeding and biotechnology.

NCP 15: Learning and inspiration

Learning and Inspiration: The provision, by landscapes, seascapes, habitats or organisms, of opportunities for the development of capacities that enable humans to thrive through education, the acquisition of knowledge and the development of skills to well-being, information and inspiration for technological art and design (for example: biomimicry): a. learning b. Apprenticeship - artistic, c. Learning - scientific and technological inspiration.

NCP 16: Physical and psychological experiences

Provision, by landscapes, seascapes, habitats or organisms, of opportunities for physically and psychologically beneficial activities (such as healing, relaxation, recreation, leisure, tourism and aesthetic enjoyment), based on close contact with nature. For example: hiking, recreational hunting and fishing, bird watching, snorkeling, gardening.

NCP 17: Supporting identities

Landscapes, seascapes, habitats or organisms that are the basis of religious, spiritual and social cohesion experiences. Provision of opportunities by nature for people to develop a sense of place, purpose, belonging, rootedness or connection, associated with different entities of the living world (for example: cultural, sacred and heritage landscapes, sounds, smells and sights associated with experiences of childhood, iconic animals, trees or flowers). Basis for narratives and myths, rituals and celebrations provided by landscapes, seascapes, habitats, species or organisms. Source of satisfaction derived from knowing that a certain landscape, seascape, habitat or species exists in the present.

NCP 18: Maintenance of options

Capacity of ecosystems, habitats, species or genotypes to keep human options open to support a good quality of life later.

- **Survey:**

The survey itself is a checklist of the NCP, where the surveyed marks each NCP that they think is related to community resilience to tsunamis.

Finally, they could provide a justification for the answer. Examples bellow:

“Función restauradora de paisajes tras una crisis socio-natural (apego al lugar, etc.)”, i.e., landscapes have a restorer function after socio-natural crisis, for example, due to place attachment.

“Se marcan los NCP que dicen en relación (1) a mantener los ecosistemas relacionados a los cuerpos de agua y de suelo en relación al evento tsunami (agua, costas, borde costero) y (2) a la relación que tienen los anteriores con el ser humano en relación a su ocupación. Es decir, cómo el ser humano se reconoce dentro del macro ecosistema, su rol, y el de la naturaleza en esta relación, que debiese ser simbiótica”, i.e., checked NCPs related to (1) water bodies maintenance ecosystems after tsunamis and (2) human role in this ecosystems and our relation with nature, that should be a symbiotic relationship.

Appendix D. Total Classes Obtained for the 50 Cities in 2021

Table A2. Total occurrences per class.

Class	Occurrences	Percent
Aves	170,933	97.2%
Magnoliopsida + Liliopsida + Polypodiopsida + Pinopsida + Gnetopsida	2689	1.5%
Insecta + Malacostraca + Arachnida + Maxillopoda + Diplopoda	948	0.5%
Reptilia + Mammalia + Elasmobranchii + Amphibia + Holocephali + Ascidiacea + Actinopterygii	353	0.2%
Gastropoda + Bivalvia + Polyplacophora + Cephalopoda	340	0.2%
Polychaeta + Clitellata + other	145	0.1%
Agaricomycetes + Tremellomycetes + Dacrymycetes	124	0.1%
Anthozoa + Scyphozoa + Hydrozoa	101	0.1%
Asteroidea + Echinodea + Holothuroidea	94	0.1%
Phaeophyceae	30	0.017%
Florideophyceae	21	0.012%
Ulvophyceae	12	0.007%
Dothideomycetes + Lecanoromycetes + Sordariomycetes + Pezizomycetes	8	0.005%
Not determined	6	0.003%
Marchantiopsida + Jungermanniopsida	5	0.003%
Hoplonemertea + other	5	0.003%
Gymnolaemata	3	0.002%
Myxomycetes	2	0.001%

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